SOUVIK ROY

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PROFESSIONAL SUMMARY

Firmware engineer building and debugging enterprise storage controller firmware in embedded C on ARM Cortex-R5. Deliver features and reliability fixes across NVMe, PCIe, SATA, NAND flash, SED/TCG key management, Redfish and PLDM, shipped into tier-1 OEM server firmware branches. Strongest at root-causing intermittent, hardware-dependent defects from UART logs and source-level debugging — including release-gate MustFix items.

PROFESSIONAL EXPERIENCE

Wipro Limited — Project Engineer *(Software/Firmware Engineer)* | Pune, India | Feb 2026 – Present

*Deputed to **Marvell Technology** since Mar 2026 — Flash R&D, enterprise storage controller firmware · embedded C, ARM Cortex-R5*

- Ship firmware features and reliability fixes across five storage subsystems (SED/TCG key management, volume state, storage event logging, Redfish and hot-plug), including ECDSA verification for PLDM images, merged into multiple tier-1 OEM branches.
- Traced a spurious volume-offline-cleared event after key-manager rekey and cold boot, a defect that had survived an earlier fix, and resolved it by gating event emission on verified volume and drive state.
- Cleared a must Fix release-gate defect for a June 2026 OEM milestone by restoring the missing drive-removal event log in the non-NVMe hot-unplug path of the GPIO interrupt handler.
- Fixed an ARM unaligned-access fault in the SED predictive-failure path with a surgical, low-risk change late in a release cycle.
- Validate releases on live OEM servers: firmware flashing, hot-plug testing, SED/non-SED drive matrices and regression cycles via BMC (Redfish/IPMI), Linux/SSH and a Raspberry Pi UART console setup.

TECHNICAL SKILLS

Firmware & Storage: Embedded C, ARM Cortex-R5, bare-metal firmware, bootloader, embedded systems, interrupt handling, GPIO, memory-mapped I/O, hardware bring-up, firmware flashing

Storage & Protocols: NVMe, PCIe, SATA, NAND flash, SSD/storage controllers, SED (TCG Opal), Redfish, PLDM, ECDSA image signing, storage event logging, hot-plug

Debug & Validation: JTAG (ARM DSTREAM), UART log analysis, source-level debugging, gdb, root-cause analysis, firmware validation, regression testing, OEM BMC interfaces (Redfish/IPMI), Minicom

Systems & Networking: Linux, POSIX threads, TCP/IP sockets, IPC, synchronization primitives, SSH

Languages & Tools: C, C++17, Python, Java, Git, JIRA, Postman, Raspberry Pi

PROJECT WORK

Enterprise Concurrent Signal Broadcast Framework — *C++17, POSIX threads, TCP/IP sockets, Linux* **2026** GitHubLink

- Built a multi-threaded Linux TCP server that fans out messages to concurrent clients, using a dedicated thread per session and mutex-guarded access to a shared subscriber registry.
- Handled session lifecycle end to end — connect, authenticate, disconnect and abrupt client loss — with deterministic cleanup and no leaked descriptors or orphaned threads on shutdown.

TRAINING

Linux Systems Programming — Wipro Limited | Feb – Mar 2026

EDUCATION

University of Engineering & Management, Kolkata — *B.Tech — Computer Science & Engineering* **2021 – 2025 | CGPA: 8.3**

Pannalal Institution — *Higher Secondary — WBCHSE* **2021 | 82%**

Bedibhawan Rabitirtha Vidyalaya — *Secondary — WBBSE* **2019 | 84%**